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URBAN HEAT ISLAND AND THERMAL COMFORT ANALYSIS USING LST, ATI AND LAND COVER CLASSIFICATION OF BENGALURU DISTRICT

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INTRODUCTION

More than half of the population these days are living in urban areas. Projections suggest that by 2050, 68.4% of the global population or more than two-thirds of all people will be living in urban areas. As of 2024, it is estimated that over **2 billion people** in urban areas experienced at least 30 "risky heat days". India is particularly vulnerable due to its tropical climate and rapid unplanned urbanization. The "concrete jungle" effect is turning Indian metros into heat traps. Bengaluru (formerly the "Garden City") offers the most startling statistics on how rapid urbanization drives local temperature rise. Vulnerable hotspots in the district includes Peenya Industrial area and Shibajinagar commercial street

Urban heat islands (UHI's) occur when urban areas become significantly warmer than their rural surroundings due to human activities. UHIs is a consequence of the greater absorption of electromagnetic energy and the slow cooling of urbanized surfaces compared to surrounding areas with the presence of vegetation (Imhoff, Marc L., et al,2010). This can happen due to urbanization, population pressure, industrialization, vegetation insufficiency, and transportation system (Halder., et al,2021). Main contributing factors are **urban surfaces** like concrete which reflect less sunlight compared to natural features**, urban geometries** which can contribute to **urban canyon effect** which led to trapping of heat near ground surfaces, next important factor is anthropogenic effect, like driving cars, air conditioning, as urban atmosphere contains more pollutants and high-water vapor content it often results in a phenomenon called urban greenhouse effect.

Measuring Urban heat island intensity is a task which requires scientific backing. It becomes tricky as temperature difference vary widely both spatially and seasonally. It can be measured as surface temperature, air temperature or by combing both. Air temperature reflect condition experienced by people while surface temperature is the heat energy given off by the ground. In 2003 Voogt and Oke suggested Surface Urban Heat Island (SUHI), this in addition to studying surface temperature differences between urban and rural areas, also addresses temporal variability.

Recent technological advancements in temperature monitoring have greatly increased current understanding of heat islands and their distributional consequences. Traditionally, urban heat islands were measured by taking the difference in temperature between the city canter and surrounding rural areas as measured by ground-based temperature monitors. In the last several decades, satellite-derived measures of air temperature have allowed for continuous mapping of the heat island effect across urban landscapes. Most recently, distributed monitor networks and community science have allowed for even more detailed mappings, such as those produced by the National Oceanic and Atmospheric Administration's Urban Heat Mapping Campaigns, which provide high resolution air temperature and humidity data throughout the day that can be used for resilience planning and research.

Land surface temperature, also known as skin temperature, is a crucial tool for studying the UHI effect using remote sensing and GIS. By analysing Land Surface Temperature (LST)data, researchers can map and understand how urban structures and land cover changes impact local temperatures helping to identify areas with high heat stress. Satellites equipped with thermal sensors, such as Landsat and MODIS, provide LST from satellite thermal bands. Combining LST with land use and land cover data provides a comprehensive understanding of the factors contributing to the UHI effect.

THERMAL SENSORS

Thermal sensors are used to analyse LST which in turn is used to calculate UHI. Such sensors/satellites include Landsat, Moderate Resolution Imaging Spectroradiometer (MODIS), Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER). Thermal sensor utilises 8–14 µm range, which is known as atmospheric window. This range includes emission part of the earth's spectrum, making heat detection possible.

The Landsat mission began in 1972, but it was from 1980s only, with Landsat 4, thermal data began to be collected, due to the recording capabilities of the following sensors: in Landsat 4 and 5, the Thematic Mapper (TM), in Landsat 7, the Enhanced Thematic Mapper Plus (ETM+) and in Landsat 8, the Thermal Infrared Sensor (TIRS) and Operational Land Imager (OLI). The Landsat family operates in sun synchronous and near-polar orbit with a nominal altitude of 705 km, an orbital inclination of 98.2°, and a revision time of 16 days. Thematic mapper began operation first with seven bands and band 6 recording thermal data, instantaneous field of view (IFOV) in band 6 is 120m*120m and other bands it is in 30m*30m. In case of ETM+, which began operation 1999 with one thermal band and seven other bands. But it has gaps caused by scan line correction failure. Next sensor was TIRS and OLI. Below table gives information about thermal sensors of Landsat 8 and 9.

Satellite	**Sensor**	**Band**	**Wavelength (µm)**	**Spatial Resolution**	**Notes**	**Commencement of Operation**
Landsat 8	TIRS-1	Band 10	10.6–11.19	100 m (resampled to 30 m)	Primary thermal band (used for LST)	Operational since May 2013 (Launch: Feb 11, 2013)
Landsat 8	TIRS-1	Band 11	11.5–12.51	100 m (resampled to 30 m)	Affected by stray light; less reliable	Operational since May 2013

Landsat 9	TIRS-2	Band 10	10.6–11.19	100 m (resampled to 30 m)	Improved calibration; accurate thermal retrieval	Operational since January 2022 (Launch: Sept 27, 2021)
Landsat 9	TIRS-2	Band 11	11.5–12.51	100 m (resampled to 30 m)	Reliable split-window use	Operational since January 2022

Advanced Spaceborne Thermal Emission Reflection Radiometer (ASTER) is a sensor onboard NASA's Terra satellite, launched in December 1999. It collects data of three different regions of EMR, in the Visible Spectrum (VIS) and NIR region there are three spectral bands with 15 m spatial resolution (bands 1 to 3), in the shortwave infrared there are six bands with 30 m resolution (bands 4 to 9), and in the **thermal infrared** region there are five bands with 90 m resolution (bands 10 to 14)

Polar and geostationary satellites can also be used for LST estimation. Their fixed positioning at a point on earth provides wide temporal coverage of local data. However, the spatial resolution and coverage area do not present much detail, which can compromise the identification of land surfaces in a more refined way, and are not suitable for analyses of UHI patterns and dynamics about heat-associated health risk at urban scales. Some examples include Spinning Enhanced Visible InfraRed Imager (SEVIRI), Suomi National Polar-orbiting Partnership (NPP), aboard Meteosat Second Generation (MGS).

LST can also be computed by from passive microwave sensors. Such sensors sense emissions from earth's surface and can also penetrate clouds, snow, rains, vegetation complementing the data for UHI studies. But course spatial resolution and inability to get temperature from frozen surface are major hindrance in using microwave sensors. Passive sensors can be used as complementing data for thermal sensor derived information but use of passive microwave sensors is rare in academia.

For the present study MODIS is an instrument onboard Terra (1999) and Aqua (2002) satellites, both launched by NASA. It has a spectral resolution of 36 bands, divided into the visible, NIR, and infrared wavelengths. With a revisit time of every one or two days, its spatial resolution is low: bands 1 and 2 are recorded with 250 m, bands 3 to 7 with 500 m, and bands 8 to 36, with 1 km.

Sensor	Platform	Orbital Frequency	Spatial Resolution	Spectral Bands (µm)	Number Band	Data Available Since
MODIS	Terra	Twice daily	1 km (approx.)	10.78–11.28	31	1999
				11.77–12.27	32	
MODIS	Aqua	Twice daily	1 km (approx.)	10.78–11.28	31	2002
				11.77–12.27	32	

Table no 1: Thermal Satellite Sensors.

Thermal data has a resolution of 1km. In MODIS LST Collection-6 products, the MODIS Land Surface Temperature and Emissivity (LST&E) product (MOD21), the LST is calculated by the ASTER Temperature Emissivity Separation (TES) algorithm and are more sensitive to land cover changes when compared to other emissivity products.

Present study computed LST from MODIS (NASA's MOD11A2 dataset). MODIS provided high temporal resolution (twice daily), thus higher probability of acquiring cloud free data. Moreover, MODIS provides a homogenous and consistent data record from 2000 to present. Using the same sensor (MODIS/Terra) for both 2010 and 2024 eliminates inter-sensor bias and calibration uncertainties that often arise when comparing Landsat 5 (TM) with Landsat 8/9 (OLI/TIRS), ensuring that the observed increase in LST is due to physical surface changes and not sensor discrepancies.

LITERATURE REVIEW

Urban Heat Island (UHI) research has evolved substantially over the decades, with global studies increasingly highlighting the connection between land surface characteristics, thermal properties, and urban expansion. One of the earliest and most influential contributions in this field is the work of Zhang et al,. 2004, who examined the relationship between UHI intensity and thermal inertia in central Beijing using ASTER and MODIS datasets. Their findings demonstrated a strong negative correlation between apparent thermal inertia (ATI) and UHI intensity, suggesting that differences in subsurface thermal properties significantly influence spatial variations in urban surface temperature. The study also emphasized the role of land-cover heterogeneity—such as built-up roads, vegetation, and water bodies—in shaping the thermal landscape. More recently, research in Southeast Asia has expanded the understanding of UHI formation under tropical climatic regimes. For instance, Putra et al. 2023 investigated Bogor City, Indonesia, and reported a consistent rise in LST over a decade, driven primarily by rapid urbanization and the fragmentation of green spaces. Their analysis showed a strong negative association between NDVI and LST and highlighted the cooling footprint of urban parks, reinforcing the crucial role of vegetation in moderating thermal discomfort in tropical cities.

Parallel to international studies, substantial work has been carried out in India to quantify UHI patterns and understand their ecological implications. Studies across different Indian cities commonly reveal that increasing impervious urban surfaces and declining vegetation cover are the primary contributors to rising LST. In Varanasi, for example, Singh et al. (2021) observed a persistent SUHI effect closely linked to expansion of built-up areas and reduction of riparian and urban green cover. Their work demonstrated strong negative and positive correlations between LST and NDVI/NDBI respectively—an observation echoed across multiple Indian contexts. Similarly, Halder et al. (2021) examined Kolkata and adjoining regions over a 30-year period and reported a 4.2°C rise in surface temperature alongside a 33.78% increase in built-up land and a 25.21% decline in vegetation. The study concluded that LST trends are strongly coupled with land transformation driven by industrialization, urban sprawl and transportation growth.

Other Indian cities exhibit comparable patterns. In Vijayawada, Kumar et al. (2012) documented significantly higher LST in dense urban cores compared to surrounding agricultural and rural zones, with NDVI showing a robust negative correlation ($r = –0.79$) with surface temperature. Research in Skopje, Macedonia—though outside India but methodologically influential—by Koplan et al. (2022) further supports this trend, demonstrating that decreasing vegetation cover directly intensifies SUHI conditions through rising LST and expanding heat-stress zones. In the Indian context, Singh et al. (2023) reported that built-up land in Bhopal increased by more than 12% between 2008 and 2020, accompanied by a 15% reduction in vegetation cover and an approximate 4°C rise in UHI intensity. Their study emphasized the importance of geospatial modelling for monitoring urban environmental changes and formulating sustainable land-use policies.

Large metropolitan areas such as Mumbai have also been extensively studied due to their rapid and unplanned urban growth. Grover and Singh (2016) highlighted that dense built-up zones, commercial corridors and expanding suburbs exhibit the highest LST values, while coastal areas and mangrove belts maintain significantly lower temperatures due to sea-breeze influence and evapotranspiration. Similar patterns were reported by Kumar (2023) across four major cities in Andhra Pradesh, where dense commercial and industrial zones consistently formed thermal hotspots, whereas vegetation and water bodies acted as strong cooling agents. The increase in SUHI intensity over time was closely tied to declining green spaces and the spread of impervious surfaces.

This clearly indicates that UHI formation is a function of complex interactions among land use dynamics, vegetation distribution, surface composition, and thermal properties. Across all studies, NDVI emerges as a key predictor of cooling, while NDBI and built-up fractions strongly correlate with higher LST. The growing evidence underscores the need for integrating geospatial approaches in urban planning, particularly in rapidly developing cities where unchecked land conversion continues to amplify thermal stress and environmental vulnerability.

Quantifying, mapping and visualising UHI of rapidly urbanizing spaces have several applications and advantages. Such studies help city planners, governments to plan strategies to mitigate the UHI and its impacts. Policy makers can take informed decisions, as UHI studies will give them a HeadStart to identify UHI hotspots, spatial and temporal variation in UHI intensity. Present study used LST derived from MODIS of 2010 and 2024 to analysed temporal variation in Urban Heat Iland. NDVI, NDBI, ATI, LULC layers were also analysed to understand the correlation with spatial distribution of UHI in Bengaluru district.

STUDY AREA

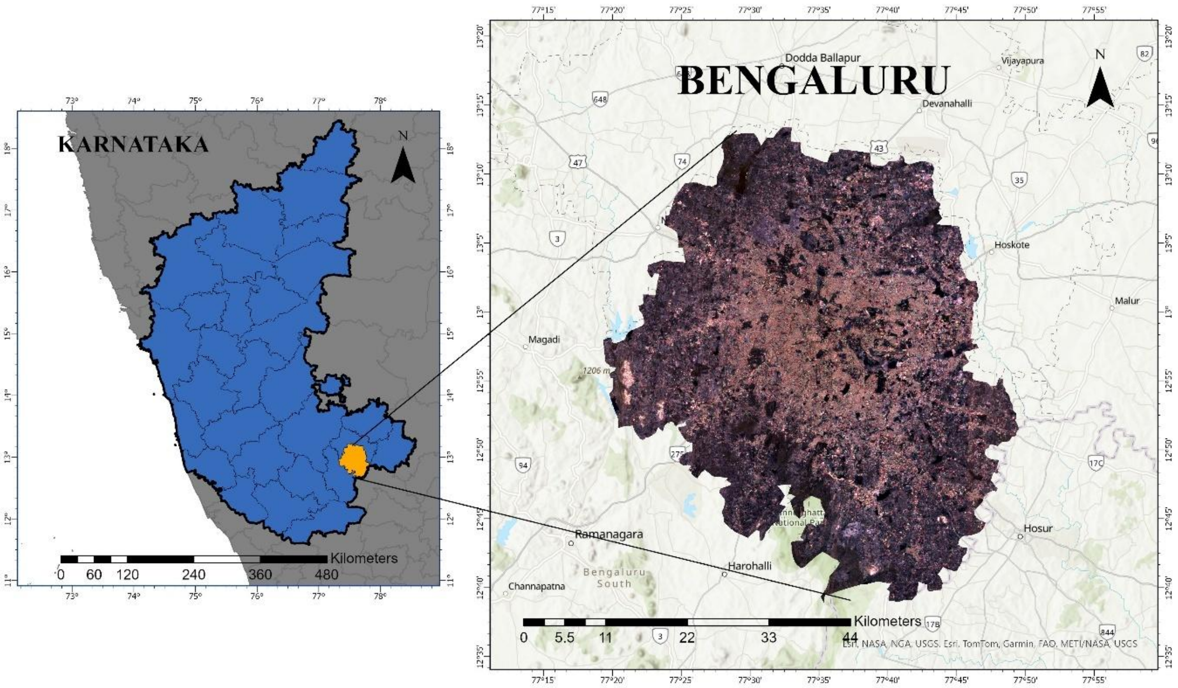


FIG 1: STUDY AREA

The capital of the southern Indian state of Karnataka, Bengaluru, is known as both the "Silicon Valley of India". The city is situated on the Deccan Plateau, over 900 meters (3,000 ft) above sea level, which gives it a milder climate compared to many other major Indian cities. Bangalore covers an area of approximately 741 sq. km and is characterized by rapid urban expansion, dense built-up zones, mixed residential–commercial land use, and shrinking green spaces and water bodies. As of recent estimates, it is the third most populous city in India with over 18 million, with a population density of 16,925 people per square kilometre. District has a diverse population due to its status as a major metropolitan and technology hub.

Bangalore experiences a tropical savanna climate (Aw) with moderate temperatures throughout the year. The district receives an average annual rainfall of around 970 mm, largely during the Southwest Monsoon (June–September) and partially from the Northeast Monsoon (October–November). Summer temperatures typically range between 28°C and 35°C, while winter temperatures vary between 15°C and 25°C. These climatic conditions, along with seasonal changes, significantly influence thermal characteristics such as Land Surface Temperature (LST) and Apparent Thermal Inertia (ATI).

Land Use and Land Cover (LULC) in Bangalore are highly heterogeneous. The central and eastern parts are dominated by dense built-up and commercial zones, while the northern and southern regions show a mix of residential, industrial, agricultural, and open/barren land. The city's physical landscape includes undulating terrain, scattered lakes, and a mixture of urban, peri-urban, and vegetated areas, making it an important region for studying urban environmental issues. Rapid urbanization, IT sector growth, and infrastructure expansion have resulted in significant reductions in vegetation cover land cover change, stress on water resources and increased impervious surfaces, contributing to urban heat island (UHI) formation. These characteristics make the city a suitable study area for applying remote sensing and GIS techniques to analyze land surface temperature (LST), land cover dynamics, apparent thermal inertia (ATI), and urban thermal comfort.

MATERIALS AND METHODS

The Surface Urban Heat Island (SUHI) Intensity was mapped by calculating the absolute temperature difference between urban pixels and a determined rural baseline temperature, classifying the city into five thermal magnitude zones (Estoque et al., 2017) for 2010 and 2023. Parallelly, the Ecological Evaluation of Thermal Comfort was assessed using the Urban Thermal Field Variance Index (UTFVI). This index normalized the LST against the regional mean temperature to delineate areas of varying ecological stress, categorizing the landscape into 3 zones ranging from comfortable to critical heat.

DATA SET USED

Sl No.	DATA SET	SATELLITE	DATA SOURCE	YEARS	RESOLUTION
1	Study area map	Sentinel 2 (GEE)	Google Earth Engine	2024	10 meters

2	LST	MODIS (MOD11A2 collection)		2010, 2015, 2020, 2024	1000 m
3	NDVI	Landsat 7 collection 2 level 2		2010	30m
		Sentinel 2(copernicus collection)		2024	10m
4	NDBI	Landsat 5 Surface Reflectance (Collection 2, Level 2)		2010	30m
		Sentinel 2		2024	10m
5	ATI	MODIS (MOD11A2 collection)		2024	1000m
6	LULC	European Space Agency's Sentinel-2-Dynamic world collection		2024	10m
		ESRI 10m annual land use		2017	10m

Table no 2: Dataset used

SOFTWARE USED

- **Arc GIS 10.8**
- **QGIS 3.38.3**
- **Google Earth Engine**

LAND USE AND LAND COVER MAPS:

To make LULC maps, Satellite images for the years 2017 and 2024 were downloaded from the Google Earth Engine using JavaScript codes.Cloud-free scenes (<30% cloud cover) were selected to ensure accurate classification. these maps were opened in ArcGIS Pro, and processes were done to prepare the imagery for fine classification of features in all the maps. The Bengaluru administrative boundary was used to clip the imagery so that analysis was limited to the study area using the ‘Clip Raster’ tool from Geoprocessing toolbox. After this, the image was classified into different features and proper symbology was done for the same. Classes included: Water Bodies, Trees / Forest, Grassland, Flooded Vegetation, Crops / Agriculture, Shrub & Scrub, Built-Up Area and Barren Land which were shown by different colours. Map elements (north arrow, scale bar, legend, and coordinates) were added using Layout View in ArcGIS Pro. The map was finally exported from the software using PNG file format for analysis and used in the final report.

NORMALISED DIFFERENCE VEGETATION INDEX MAPS:

For NDVI map preparation, Satellite imagery was taken from Google Earth Engine.

2010 and 2024 data downloaded using the JavaScript codes. Cloud-free images (<40%) cloud cover covering the entire Bengaluru Urban area were selected.
After opening the dataset in ArcGis pro,
Output values ranged between –0.3 to 0.8, which was classified into 5 vegetation categories using Manual Breaks.
A green-based colour ramp was used to represent the map and features accurately.
Dark green → Dense vegetation
Yellow → Moderate
Red → Non-vegetation
A Legend, scale bar, north arrow, and map title were added in Layout. Then the maps for both 2010 and 2024 were exported as PNG files for the final report.

NORMALISED DIFFERENCE BUILT UP INDEX MAPS:

Cloud-free satellite images for Bengaluru city were downloaded from Google Earth Engine. We used Symbology and classified into 2 classes. To interpret built-up areas clearly, the NDBI raster was classified:

(–0.7) to 0 → Non-built-up (vegetation, water, soil)
0 to 1 → Built-up (urban, concrete, settlements)
Classes were assigned distinct colours for interpretation: Built-up shpwing Red and Non-built-up showing Yellow.

A clear accurate legend, north arrow, scale bar, title, and citation were added in Layout view. Final map layouts were exported as high-resolution PNG files and added in the final report

LAND SURFACE TEMPERATURE MAPS:

Land Surface Temperature (LST) data was acquired using the MODIS/Terra Land Surface Temperature/Emissivity 8-Day L3 Global 1km (MOD11A2) product. In order to make LST maps, following steps were done.

Downloaded LST data from MODIS collection from different years (2010, 2015,2020 and 2024). These satellite data were downloaded from Google Earth Engine using JavaScript codes. These were downloaded in the file format .tif and exported to Google drive. Opened in ArcGIS wher it projected and **resampled to a spatial resolution of 30 meters** using the Nearest neighbour method. Later, Symbology was done classification of the the map into 5 classes was done, showing variations like moderately low, low, moderate, high and very high according to temperature variations from 17.4 to 54^{0}C.After making all maps, layout was done for each map along with suitable legend, north arrow, scale and title and then exported as png format for final report.

APPARENT THERMAL INETRIA:

Thermal inertia (TI) is a physical property of materials describing their impedance to temperature change. Apparent thermal inertia (ATI) is an approximation of thermal inertia calculated as one minus albedo divided by the difference between daytime and night time radiant temperatures. Regional estimates of ATI provide useful information and have a vast application.

$$\boldsymbol{ATI = 1 - A/\Delta T}$$

A= Albedo

Delta T = Diurnal temperature difference

SURFACE URBAN HEAT ISLAND INTENSITY MAPS:

The SUHI intensity was derived using the absolute urban-rural temperature difference method (Estoque et al., 2017).

1. Determination of Rural Baseline: A rural reference temperature was established by extracting the mean LST the study area by excluding built-up and waterbodies (set null tool)

2. Intensity Calculation: The SUHI intensity for each pixel was calculated as:

$$SUHI = LST(Mean) - LSTrural$$

2010 = LST rural was 34-degree Celsius

2024 =LST rural was 35 degree Celsius

THERMAL COMFORT MAP:

To assess the impact of the thermal environment on ecological quality, the Urban Thermal Field Variance Index (UTFVI) was calculated. This index normalizes the surface temperature against the regional mean to quantify thermal stress.

$$UTFVI = \frac{LST - LSTmean}{LSTmean}$$

- (("LST" + 273.15) - (LSTmean + 273.15)) / (LST mean + 273.15) -converted to kelvin

2010 LST mean	29.8
2024 LST mean	42.6

Modified UTFVI Classification was adopted here. The standard six-level UTFVI classification (Zhang et al., 2006) was reclassified into three broad categories—Comfortable, Moderate, and Critical—to delineate priority zones for urban cooling interventions."

RESULTS AND DISCUSSIONS

The Land Use/Land Cover (LULC) maps prepared for the years 2017 and 2024 clearly depict the spatial distribution and temporal changes in land categories within the study area. The comparison of these two maps indicates noticeable shifts in land use patterns driven mainly by urban expansion, population pressure, and changes in riverine and open land conditions.

Between 2017 and 2024, the built-up area increased significantly. More continuous and compact patches appear in 2024, showing the growth of residential, commercial and transportation infrastructure.Vegetation cover shows a reduction from 2017 to 2024. In 2017, vegetation patches were more widespread, but by 2024 many of these areas have been converted into built-up surfaces or barren/open land. This indicates deforestation, land clearing for development, and a decline in green buffer zones. The shrinking vegetation may also influence local microclimate and urban heat patterns. The presence of water bodies, even if reduced, continues to play a crucial role in local drainage and ecology. Open land shows fluctuations, with some areas from 2017 being absorbed into built-up categories by 2024. Where agricultural land is present, a declining trend can be observed. Some cropland areas are either converted into settlement zones or left fallow/open. This shift suggests a change in land value, urban influence, and possible reduction in agricultural activity near the study region.

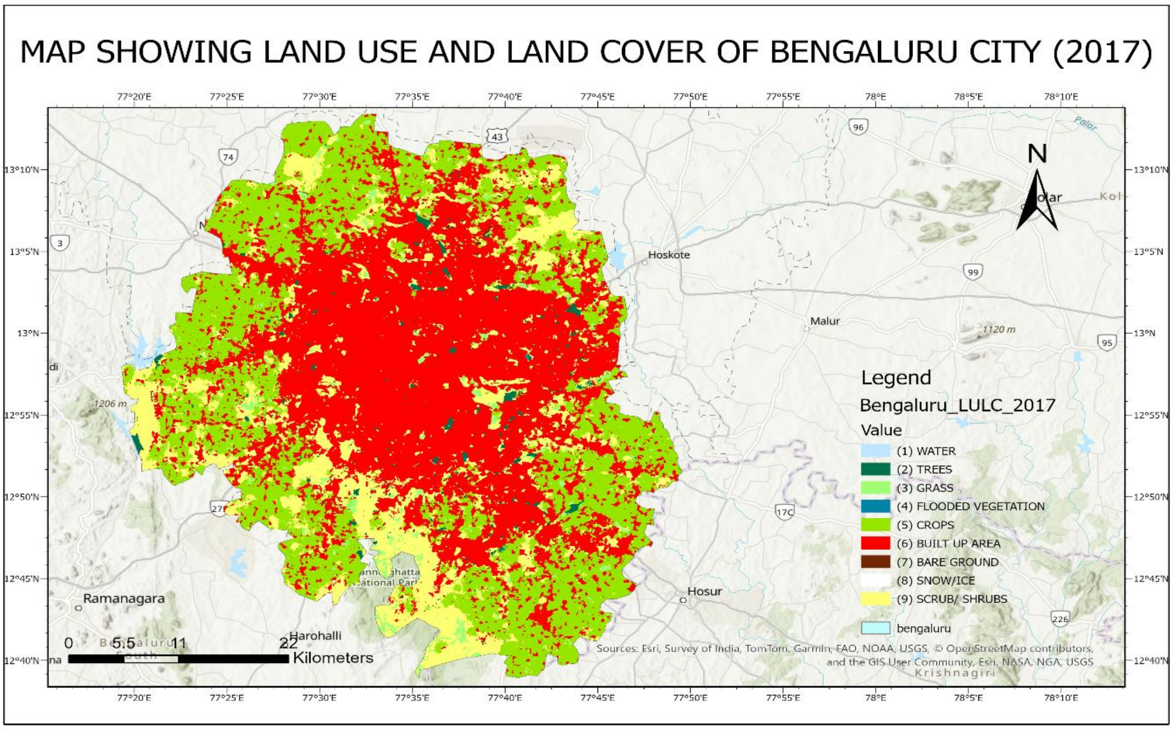


FIG NO 2: LULC MAP OF BENGALORE (2017)

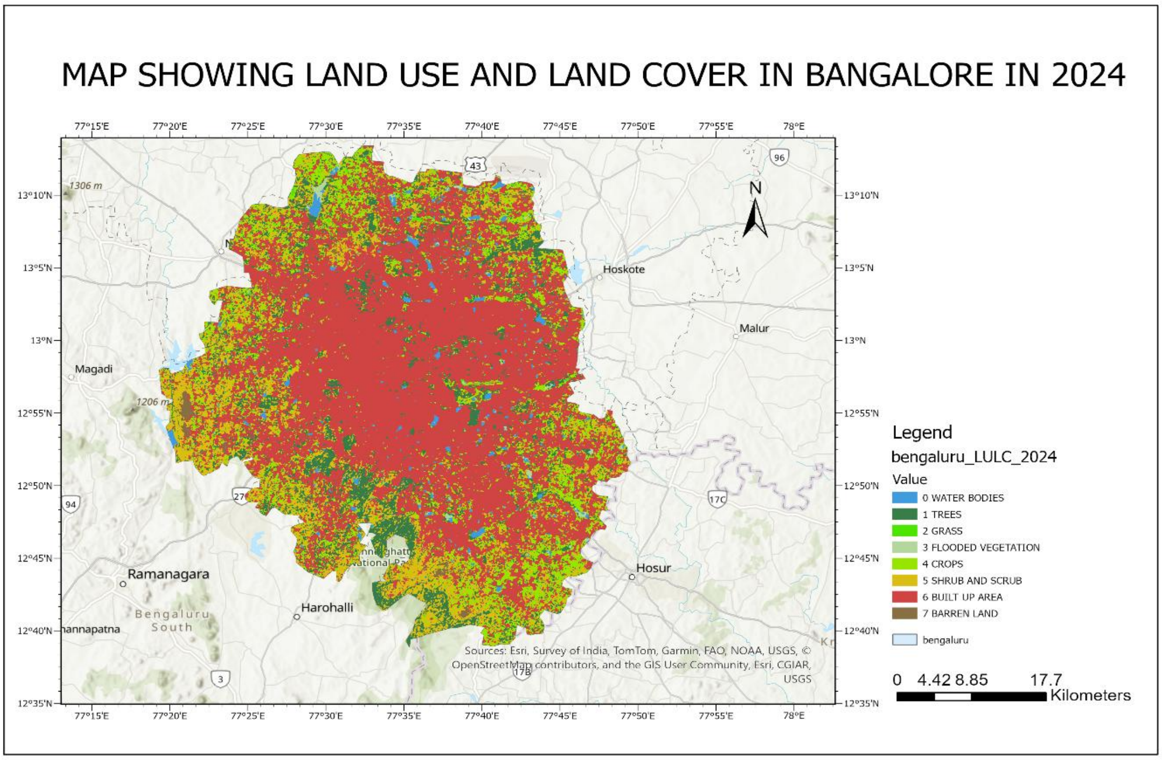


FIG NO 3: LULC MAP OF BENGALORE (2024)

The NDVI map of 2010 shows that Bangalore had a higher proportion of green cover, especially in the peripheral areas. Central Bangalore displays low NDVI values (-0.1 to 0.2), represented by red and orange colours, indicating: dense built-up areas and Roads, commercial districts Sparse vegetation. Moderate vegetation (0.2–0.4) in yellow shades is seen around residential areas and small parks and open spaces. Dense to very dense vegetation (0.4–0.8) in green shades occurs mainly in Western and southern outskirts with lesser urban sprawl.

The 2024 NDVI map shows a clear decline in vegetation. A large increase in red and orange areas (−0.1 to 0.2) marks the expansion of built-up surfaces. Moderate vegetation (0.2 - 0.4) has reduced and appears more fragmented, indicating replacement of small green patches with built structures and decrease in park areas and roadside vegetation. Overall, the 2024 NDVI reflects a significant loss of vegetation cover due to rapid urbanisation, infrastructure development, and conversion of open lands into residential and commercial zones.

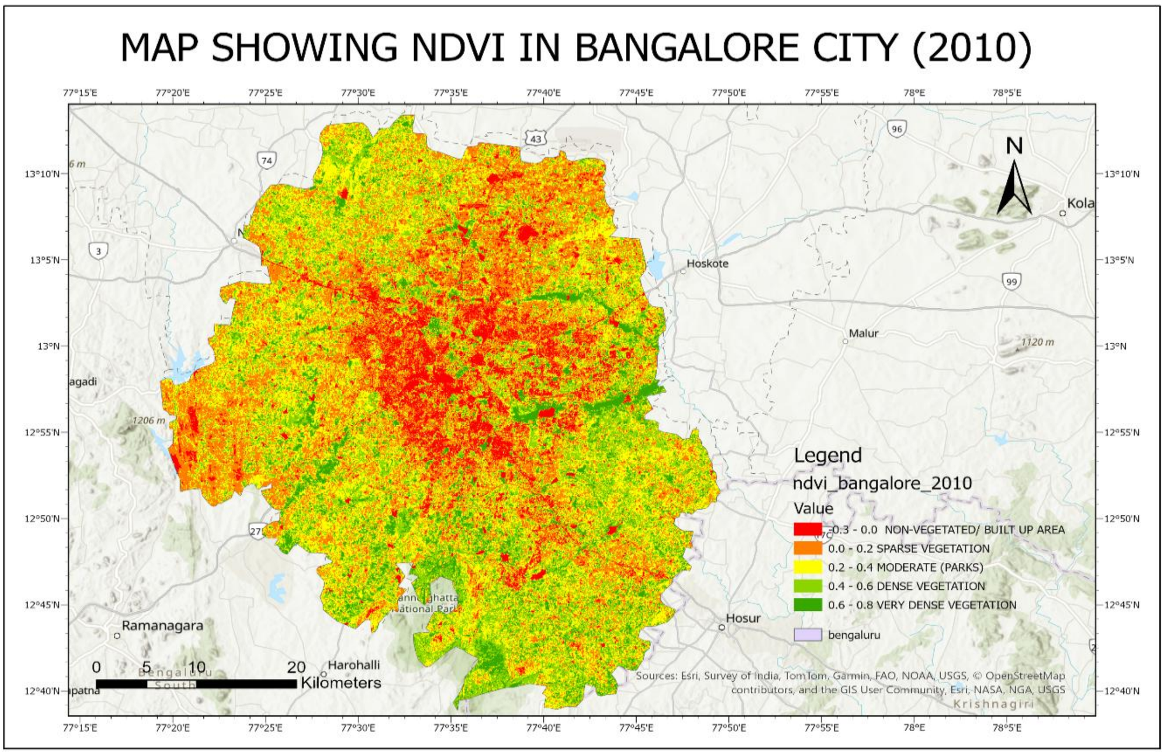


FIG NO 4: NDVI MAP OF BENGALURU (2010)

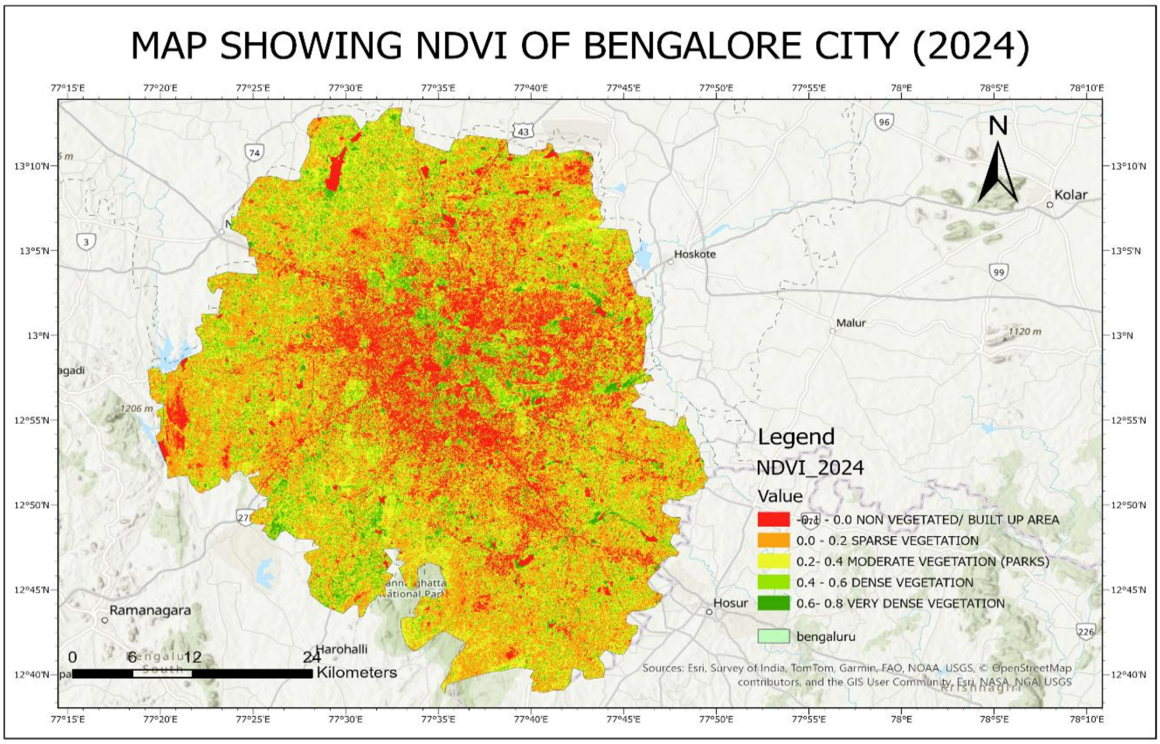


FIG NO 5: NDVI MAP OF BENGALURU (2024)

The NDBI maps for 2010 and 2024 clearly show the distribution of built-up (urban) and non-built-up areas across Bangalore city. In the map, red areas represent built-up surfaces (positive NDBI values) and yellow areas represent non-built-up surfaces (negative NDBI values).

The 2010 map shows moderate urbanization concentrated mainly around The central city region and major transportation routes. The built-up areas were dense in the core but gradually reduced toward the periphery. Some patches of open lands, vegetation, agricultural fields, and waterbodies (non-built-up areas) were still visible. This suggests that in 2010, Bangalore was still in an early-to-mid phase of spatial urban expansion and vegetative cover had started to decrease with growth of urban areas.

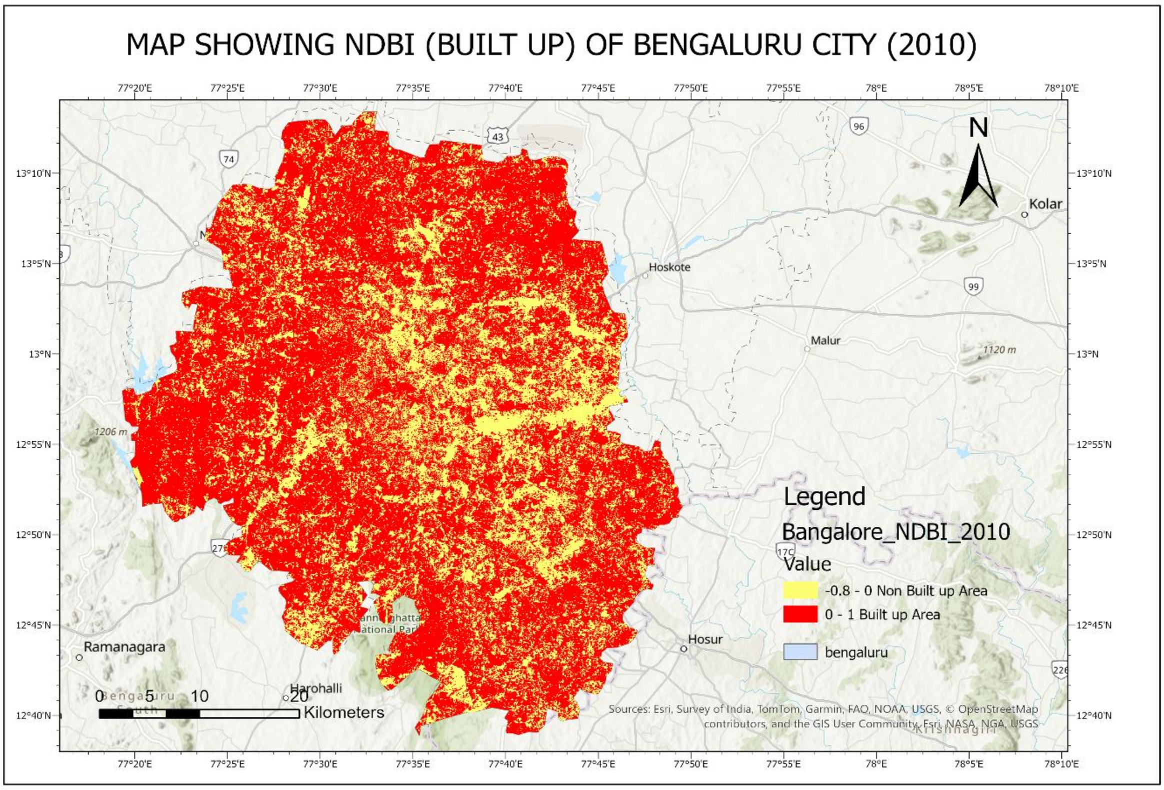


FIG NO 6: NDBI MAP OF BENGALURU (2010)

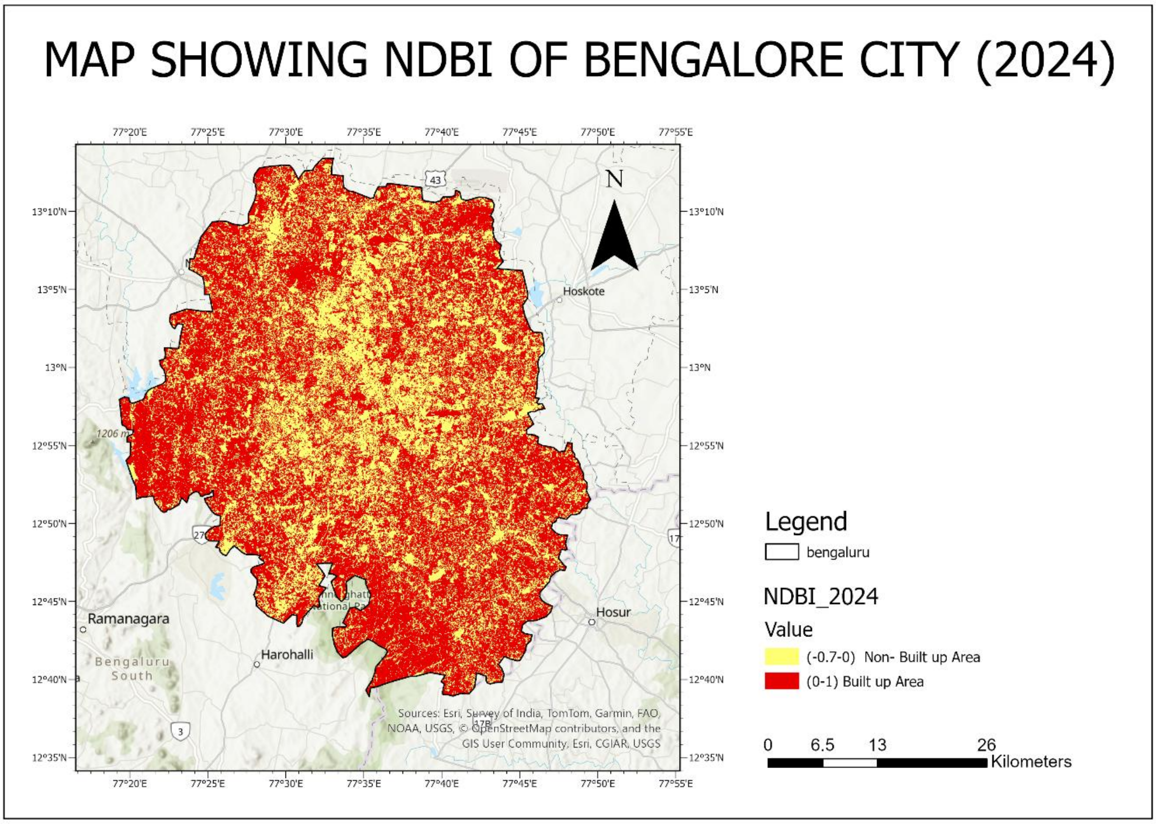


FIG NO 7: NDBI MAP OF BENGALURU (2024)

There is a significant rise in built-up surface from 2010 to 2024. The quantity of non-built-up land has drastically reduced, with very limited yellow patches remaining. The city has undergone rapid infrastructure development, leading to loss of vegetation and open land. By 2024, Built-up areas have expanded uniformly in all directions across Bangalore Metropolitan region, reducing the extent of non-built-up surfaces. The periphery, which earlier showed significant yellow (non-built-up) areas, is now almost fully urbanized showing urban sprawl and infilling.

The temporal analysis of Land Surface Temperature (LST) for Bengaluru across the years 2010, 2015, 2020, and 2024 were done and it shows an increase in temperatures throughout the years. In 2010, the majority of the city fell within moderately low (17–31°C) and low (31–34°C) temperature zones. Green colour dominated areas show vegetative cover. High Temperature areas (shown by red and orange 37–48°C) were more concentrated in high built up areas in the south-central part of the city.

In 2015, the LST shifted toward higher temperature ranges, rising to 21.71°C–54.33°C. Areas having moderately low temperature (21–33°C) decreased. Red temperature hotspots area (40–54°C) became more prominent in the south and southeastern part of the city due to rapid expansion of the built-up areas in the region where vegetation reduced and urban heat increased.

By 2020, the LST further rose to 7.71°C–53.15°C. Western and northern Bengaluru display widespread red zones, which means strong (UHI) Urban Heat Intensity development. This year marks the peak phase of heat island formation, due to increased industrial areas, real estate expansion, and reduced green spaces.

By 2024, LST starts ranging from 21°C–58°C and large portions of the central, eastern, and southern regions of the city exhibit very high temperatures (47–58°C). The green vegetative area shrinks more, now only visible in the outskirts of the Bengaluru city which indicates a severe intensification of the UHI effect.

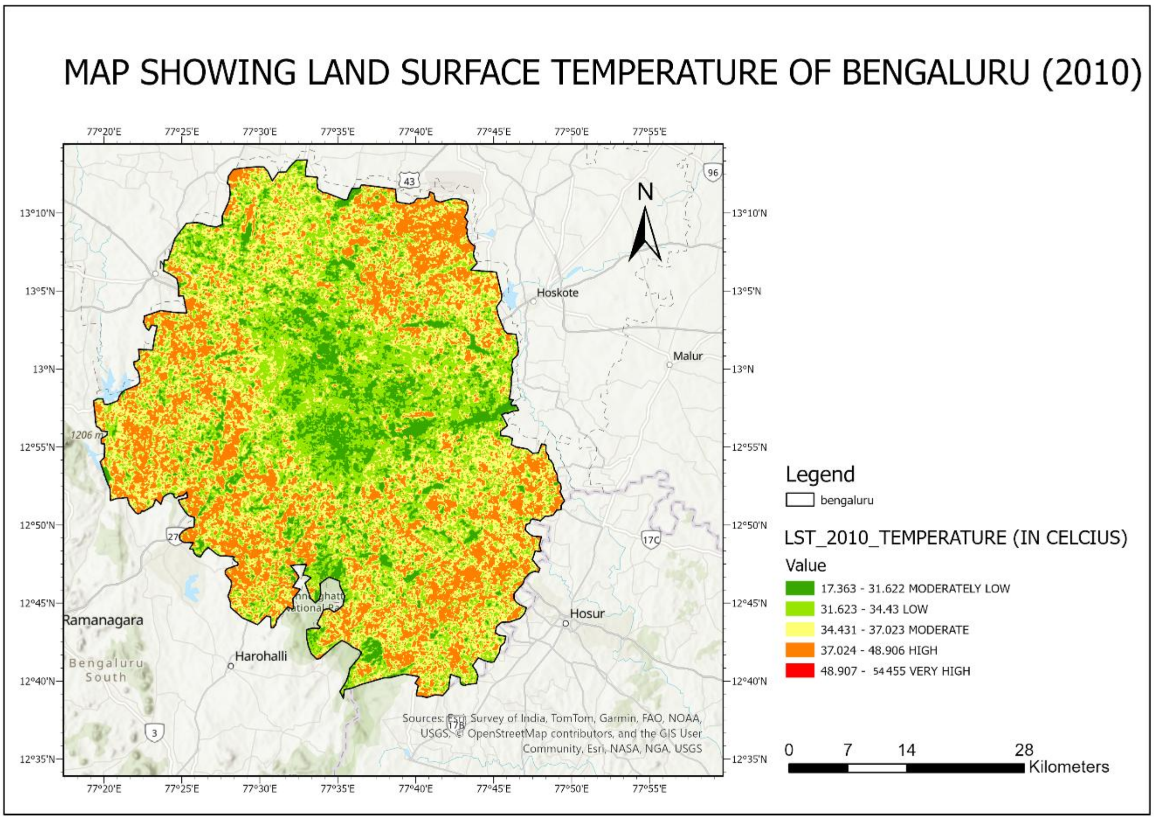


FIG NO 8: LST MAP 2010

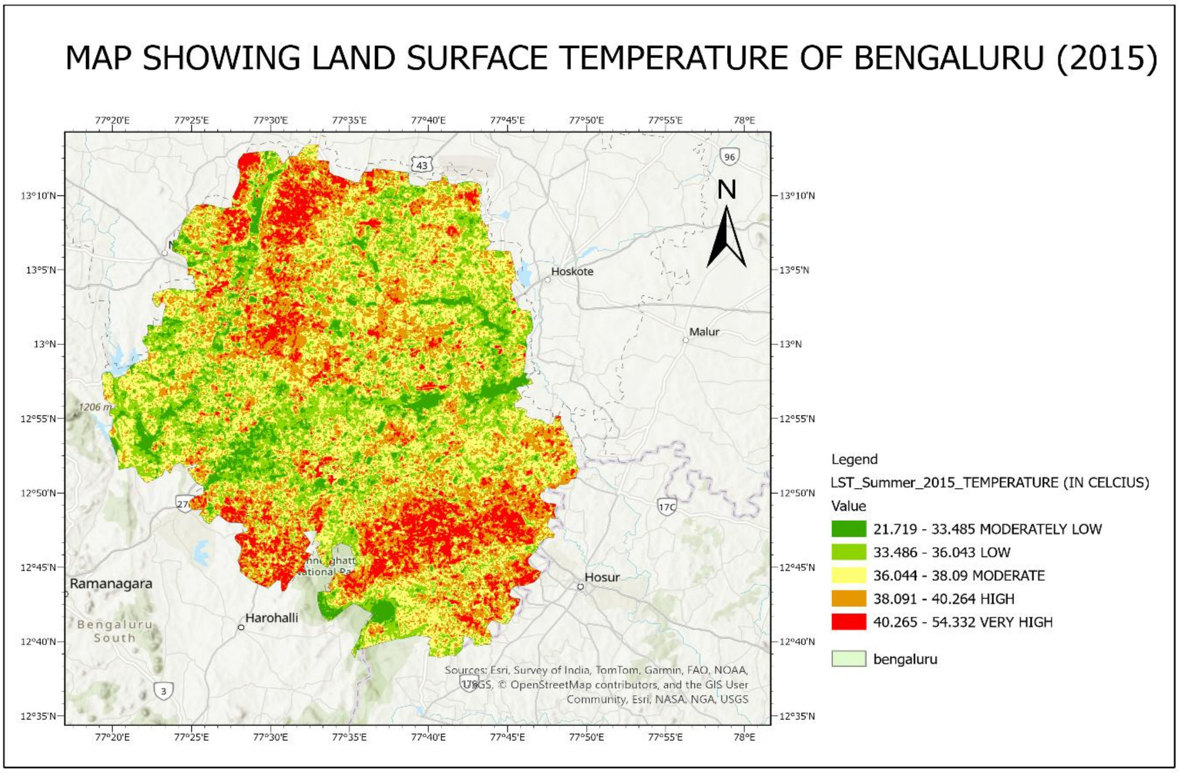


FIG 9: LST 2015

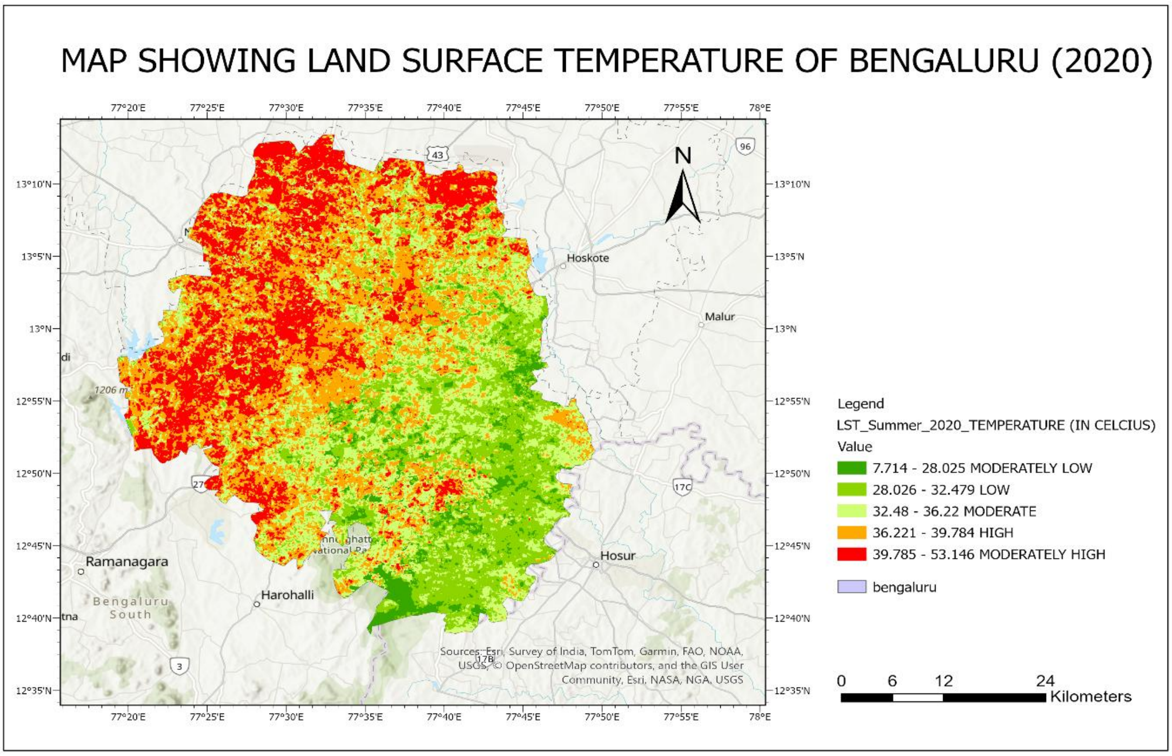


FIG 10: LST MAP 2020

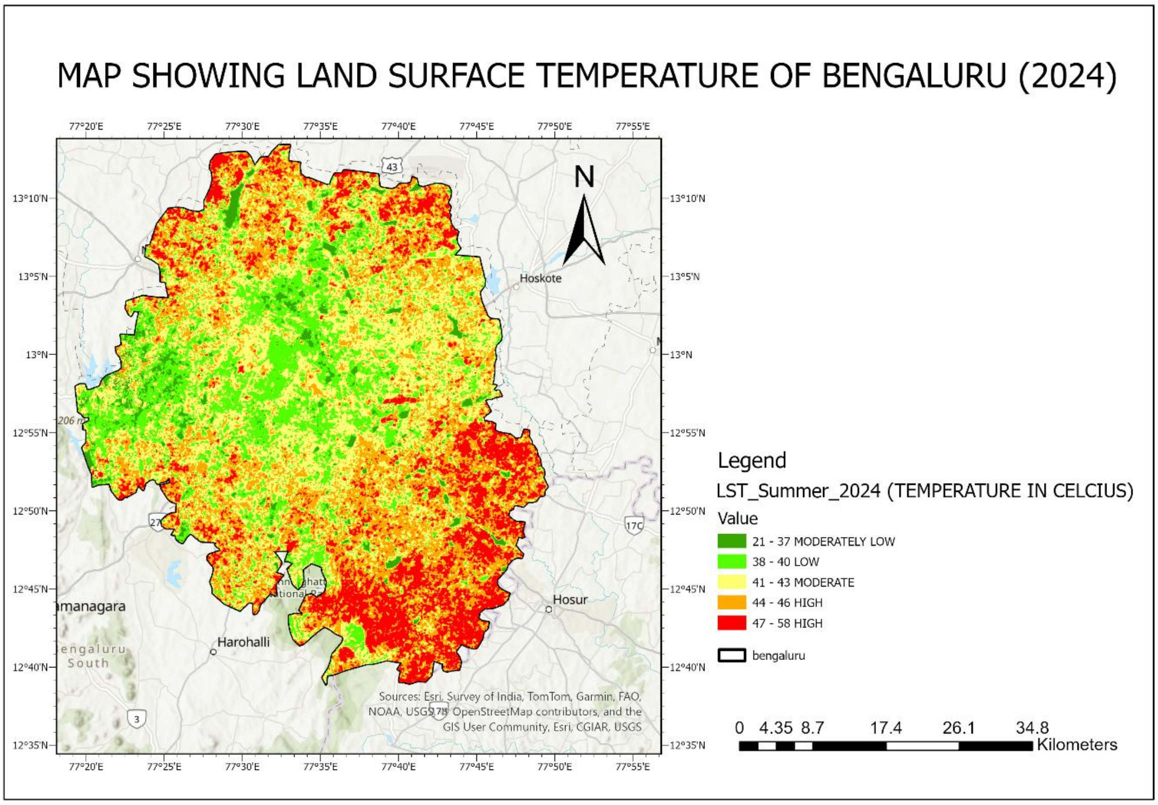


FIG 11: LST MAP 2024

Mean LST (Deg.C)	Years
29.8	2010
37.2	2015
35.8	2020
42.6	2024

Over the years mean LST increase can clearly be understood by analysing LST of above years.

Spatiotemporal Variations in Land Surface Temperature (LST) The temporal analysis reveals a marked intensification of the urban thermal environment between 2010 and 2024. The mean Land Surface Temperature of the study area rose significantly from 37.2°C in 2010 to 42.6°C in 2024, representing a net regional warming of 5.4°Cover the 14-year period.

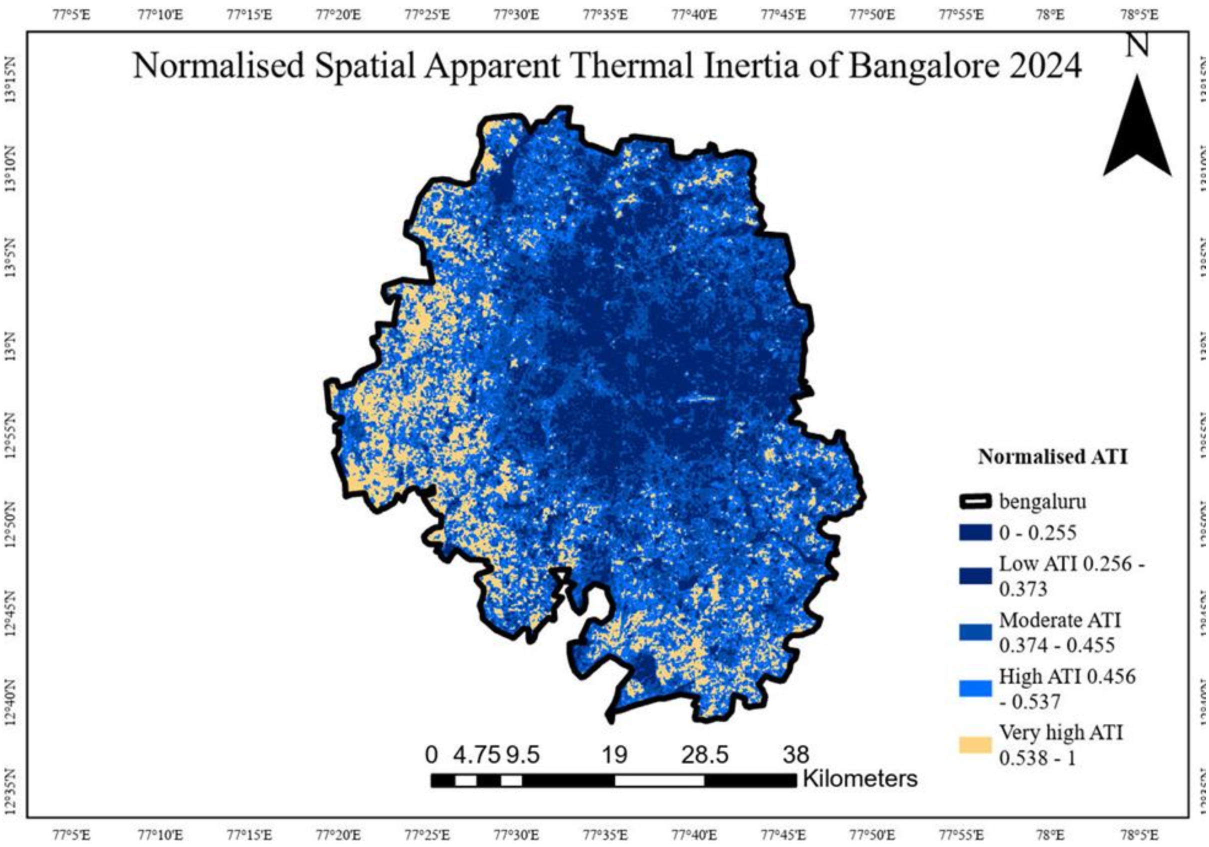


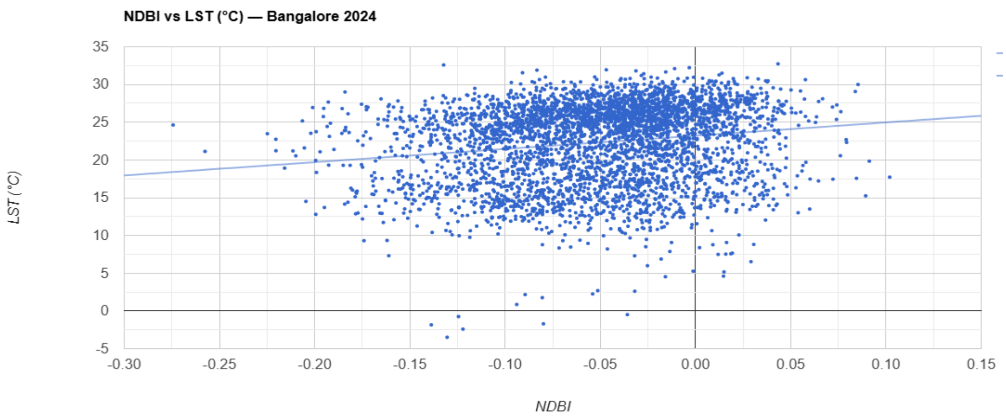
FIG 12: ATI MAP 2024

The Normalised Apparent Thermal Inertia (ATI) map of Bangalore for 2024 shows that the city is predominantly characterised by low to moderate ATI values, indicating that most land surfaces heat up and cool down rapidly. These low ATI zones, which occupy the majority of the area, are associated with dense built-up regions, paved surfaces, and barren or dry land, reflecting the high level of urbanisation across the city. Moderate ATI values appear in scattered patches and correspond to mixed urban–vegetated environments, such as partially green residential areas and tree-lined streets. In contrast, high and very high ATI values occur only in small, isolated clusters, mainly representing parks, vegetation-rich pockets, and moisture-retaining surfaces. The limited extent of these high-ATI zones suggests a fragmented green network within Bangalore. Overall, the spatial distribution indicates that Bangalore's surface characteristics are dominated by low thermal inertia materials, making the city more susceptible to daytime heating and enhanced urban heat island effects.

CORRELATION AND STATISTICAL ANALYSIS

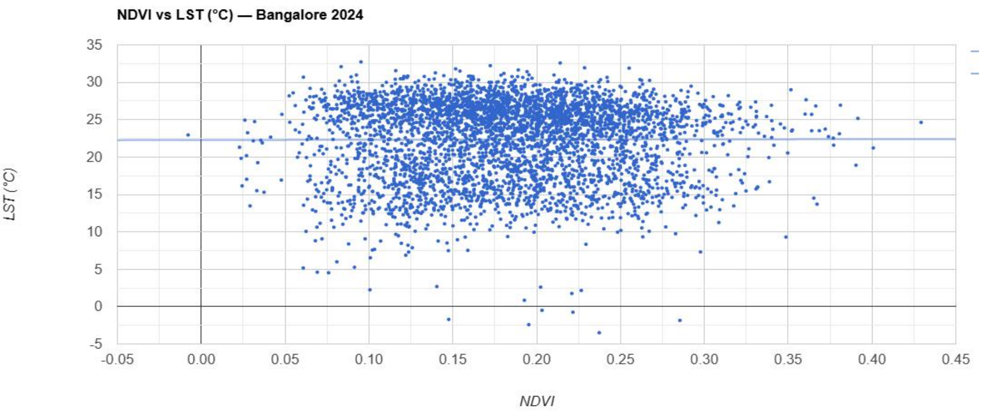
The scatter plot shows a positive relationship between NDBI and LST, meaning that as built-up intensity increases, land surface temperature also rises. Areas with higher NDBI values (more urban/built-up surfaces) generally show higher LST, indicating stronger urban heat island effects. Low NDBI regions (vegetation/open areas) have comparatively lower temperatures. This suggests that urbanization significantly contributes to surface heating in the study area.

SCATTER PLOT (NDBI vs LST 2024)



The scatter plot between NDVI and LST shows a weak relationship **($R^2 = 0.01$)** between NDVI and Land Surface Temperature (LST) across Bangalore in 2024. NDVI values remain low (0.05–0.35), indicating limited vegetation cover in much of the study area. LST mostly ranges between 20–30°C. The regression line (LST = 0.284 × NDVI + 22.277) shows a very weak positive trend, meaning NDVI does not strongly influence LST in this mixed urban landscape. The wide spread of points suggests that factors other than vegetation—such as built-up density, surface materials, and urban activity—play a larger role in controlling LST. Overall, the plot indicates that vegetation presence is limited and has only a minimal impact on temperature variation in the study area.

SCATTER PLOT (NDVI vs LST 2024)



The SUHI intensity, quantified using the urban-rural difference method, highlights a critical shift in thermal regimes.

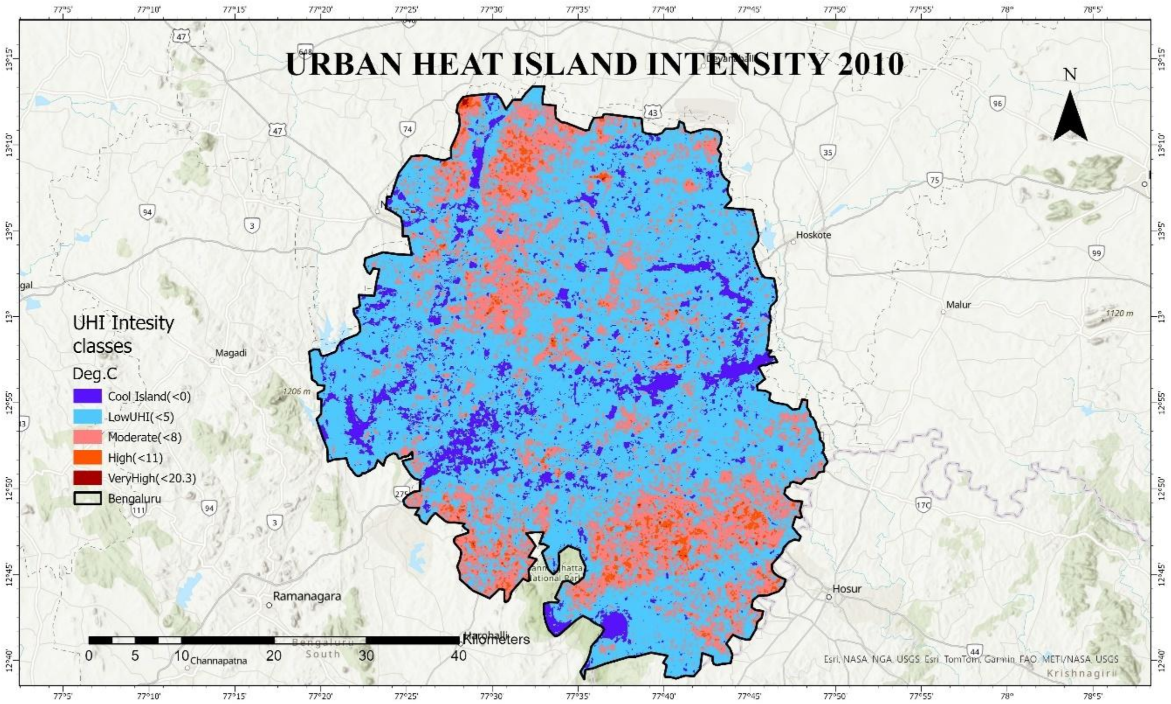


FIG 13: UHI 2010

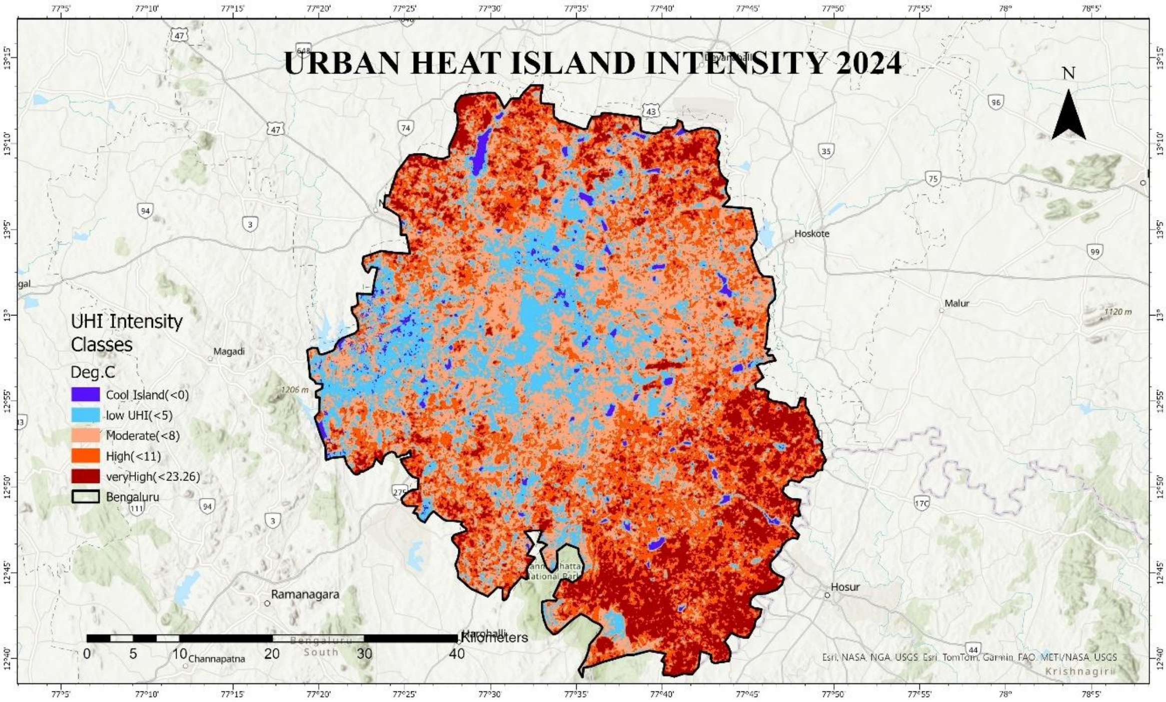


FIG 14: UHI 2024

'Extreme UHI' (> 6°C), are not confined to a single city canter but form a polycentric pattern. Industrial cluster and high-density core like CDM, Majestic etc indicated Very UHI. Cool island was spread sporadically across the region indicating presence of water bodies and urban green spaces. Such areas help in buffering effect of Urban heat in the district. Such areas include hydrological sinks like **Bellandur Lake, Ulsoor Lake, and Sankey Tank** and dense canopy zones like **Indian Institute of Science (IISc) campus, Lalbagh, and Cubbon Park**.

UHI Class 2010	**Code**	**Pixel Count**	**Area (km2)**	**Percentage (%)**	**Interpretation**
Urban Cool Island (< 0°C)	1	225,676	203.11	9.28%	Significant presence of lakes and dense vegetation.
Weak UHI (0 – 2°C)	2	1,608,247	1,447.42	66.10%	The city was mostly ‘neutral’ or mild suburbs.

Moderate UHI (2 – 4°C)	3	541,166	487.05	22.24%	Emerging residential clusters.
Strong UHI (4 – 6°C)	4	56,256	50.63	2.31%	Very few hot spots
Extreme UHI (> 6°C)	5	1,667	1.50	0.07%	Almost non-existent (Industrial areas were smaller).

UHI Class 2024	**Pixel Count**	**Area (km2)**	**Percentage (%)**	**Interpretation**
Urban Cool Island (< 0°C)	41,731	37.56	1.67%	Lakes, Parks (e.g., Ulsoor, Lalbagh)
Weak UHI (0 – 2°C)	444,070	399.66	17.73%	Suburbs, sparse vegetation
Moderate UHI (2 – 4°C)	888,438	799.59	35.46%	Residential zones, mixed-use
Strong UHI (4 – 6°C)	733,647	660.28	29.28%	High-density apartments, roads
Extreme UHI (> 6°C)	397,309	357.58	15.86%	Industrial (Peenya), CBD, Metal roofs

Expansion of Heat Stress: In 2010, the cityscape has more 'Weak UHI' (0–2°C), which covered 66.10% of the area. By 2024, high-intensity class has major share of area. The 'Extreme UHI' zone, which was virtually non-existent in 2010 (0.07%), expanded exponentially to cover 15.86% of the city in 2024. These hotspots are spatially correlated with industrial clusters like Peenya and high-density commercial business districts.

Analysis of the UHI intensity reveals that nearly 45% of the study area falls under high thermal stress categories. Specifically, 29.28% of the city experiences Strong UHI (4–6°C hotter than rural baseline), while 15.86% suffers from Extreme UHI (>6°C), primarily concentrated in industrial clusters like Peenya and dense commercial hubs. In contrast, the 'Urban Cool Island' effect, provided by water bodies and dense canopy, covers only 1.67% (37.56 km2) of the landscape, highlighting a critical deficit in thermal buffering capacity within the urban core. The dominant thermal class is 'Moderate UHI' (35.46%), corresponding to the vast residential layout of Bengaluru.

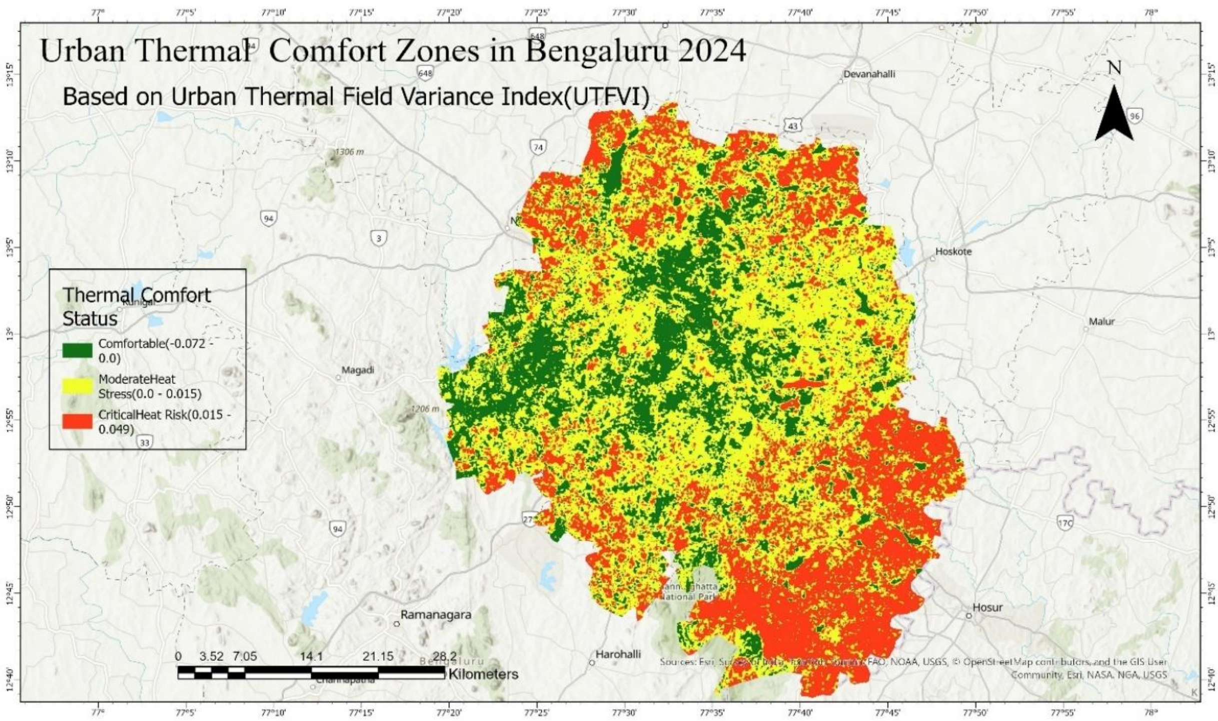


FIG 15: THERMAL COMFORT MAP 2024

Urban Thermal comfort zones indicate ecologically stressed regions in the region. It is mostly concentrated in urban areas and its spatial distribution correlated with SUHI classes very well. Comfortable zones indicate less heat stress due to green spaces and water bides in the study region, while moderate zones are coming acting as buffer from high heat stress region and comfortable regions.

CONCLUSION

Present study utilised thermal sensors and satellite data to quantify, analyse and visualise spatial and temporal Land surface temperature changes over the years. It was further used to understand the increase of Urban Heat Island effect in Bengaluru city from 2010 to 2025.

Surface Urban Heat Island intensity (SUHI) was calculated using Urban-Rural Temperature difference method. Intensity was calculated by subtracting rural mean temperature from mean LST temperature. SUHI classes indicated spatial spread urban heat island effect in the district. Comparing it with two years enabled the study of temporal changes that happened in the district. Increased urbanisation, which ultimately led to increased build-up, more concrete and asphalt surfaces and reducing green spaces have contributed to increase in Urban Heat Island effect in the region.

In 2010 only 0,07% of the district had extreme heat while it increased to nearly 16% in 2024, indicating clear increase in SUHI. Likewise, ‘cool island’ class occupied aroun10% of the district in 2010 but it reduced to around 2% by 2024. Rapid urbanisation has encroached on regions cool islands during the study period. Moreover, it can be understood that there is a shift in the dominant class of SUHI. in 2010 it was ‘Weak UHI’ (60%) but by 2024 dominant class was moderate (35%) and strong (29%) UHI. This clearly indicates that the entire thermal baseline of the city has shifted. What used to be a 'mild' city is now predominantly a 'stressed' city.

Urban heat island is an increasingly concerning phenomenon. It becomes relevant in the backdrop of urbanisation and climate change. Every year, the proportion of people exposed to heat stress is increasing. This calls for holistic mitigation strategies worldwide and especially in the Indian context with its rising population and urbanization.

This study highlights the relevance of remote sensing data to estimate and visualize UHI. Also identify factors contributing to increased UHI, this will help policy makers to formulate mitigation strategies and plan smart cities, which will contribute a sustainable and resilient future.

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